Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The temperature at which proton in hydrogen gas would have enough energy to overcome a barrier of 4.14×10^{-14} J is (Boltzmann constant $= 1.38 \times 10^{-23} \mathrm{JK}^{-1}$)

(A) 2×10^{9} K

(B) 10^{9} K

(C) 6×10^{9} K

(D) 3×10^{9} K

Q.2

The root mean square velocity of hydrogen molecules at 300 K is $1930 \mathrm{metre/sec}$. Then what is the r.m.s velocity (in $\mathrm{m/sec}$) of oxygen molecules at 1200 K ?

(A) 955

(B) 965

(C) 975

(D) 985

Q.3

N molecules, each of mass m, of gas A and 2 N molecules, each of mass 2 m, of gas B are contained in the same vessel which is maintained at a temperature T. The mean square velocity of molecules of B type is denoted by V_2 and the mean square velocity of A type is denoted by V_1, then what will be the value of $\frac{\mathrm{V}_1}{\mathrm{V}_2}$?

(A) 1

(B) 2

(C) 3

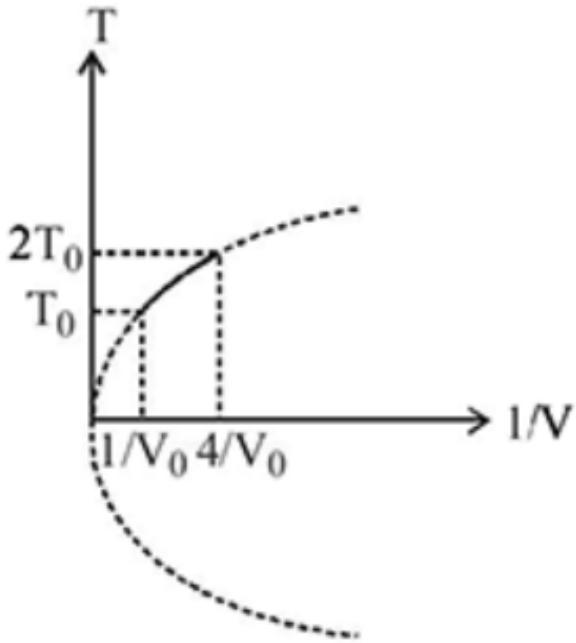
(D) 4

Q.4

Figure shows a parabolic graph between T and $1/\mathrm{V}$ for a mixture of a gases undergoing an adiabatic process.

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What is the ratio of V_{rms} of molecules and speed of sound in mixture?



(A) 1.441

(B) 1.414

(C) 1.144

(D) 1.14

Q.5

The velocity of sound in air is $332\ \text{m s}^{-1}$ at NTP. Find the rms speed of air molecules at NTP. $(\gamma = 1.41)$

(A) $484\ \text{ms}^{-1}$

(B) $418\ \text{ms}^{-1}$

(C) $248\ \text{ms}^{-1}$

(D) $382\ \text{ms}^{-1}$

Q.6

Four molecules have speeds $2\ \text{km/sec}, 3\ \text{km/sec}, 4\ \text{km/sec}$ and $5\ \text{km/sec}$. The root mean square speed of these molecules (in km/sec) is

(A) $\sqrt{54/2}$

(B) $\sqrt{54/4}$

(C) 3.5

(D) $3\sqrt{3}$

Q.7

The absolute gas temperature at which the root mean square speed of helium molecules exceeds their most probable speed by $200\ \text{m/s}$ is

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(A) $110.2\ \mathrm{K}$

(B) $90.2\ \mathrm{K}$

(C) $190.2\ \mathrm{K}$

(D) $100.2\ \mathrm{K}$

Q.8

$N(<100)$ molecules of a gas have velocities 1,2 $3, \ldots \ldots \ldots \mathrm{Nkm/s}$ respectively. Then ratio of rms speed and average speed is

(A) 1

(B) $\frac{\sqrt{(2\,\mathrm{N}+1)(\mathrm{N}+1)}}{6\,\mathrm{N}}$

(C) $\frac{\sqrt{(2\,\mathrm{N}+1)(\mathrm{N}+1)}}{6}$

(D) $2\sqrt{\frac{2\,\mathrm{N}+1}{6(\,\mathrm{N}+1)}}$

Q.9

If the molecules in a tank of hydrogen have the same RMS speed as the molecules in another tank of oxygen, we may be sure that

(A) the pressures are the same

(B) the hydrogen is at the higher temperature

(C) the temperatures are the same

(D) the oxygen is at the higher temperature

Q.10

At room temperature a diatomic gas is found to have an r.m.s. speed of $1930\ \mathrm{ms}^{-1}$. The gas is:

(A) H_2

(B) Cl_2

(C) O_2

(D) F_2

Q.11

The pressure of a gas is raised from $27^{\circ}\mathrm{C}$ to $927^{\circ}\mathrm{C}$. The root mean square speed is

(A) $\sqrt{(927/27)}$ times the earlier value

(B) remain the same

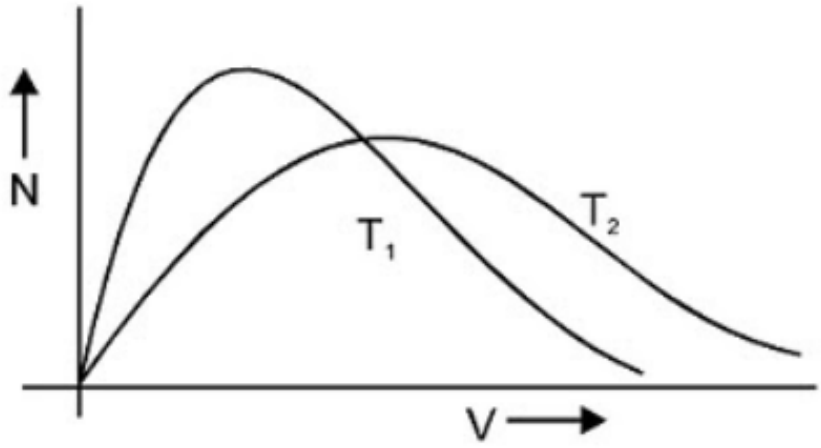
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(C) gets halved

(D) get doubled

Q.12

Maxwell's velocity distribution curve is given for two different temperature. For the given curves.



(A) $T_1 > T_2$

(B) $T_1 < T_2$

(C) $T_1 \leq T_2$

(D) $T_1 = T_2$

Q.13

The temperature of an ideal gas is increased from $120\ K$ to $480\ K$. If at $120\ K$ the root-mean-square velocity of the gas molecules is v, at $480\ K$ it becomes

(A) $4v$

(B) $2v$

(C) $v/2$

(D) $v/4$

Q.14

A nitrogen molecule has some rms speed at $0^\circ C$ on the surface of the earth. With this speed, it goes straight up. If there is no collisions with other molecules, the molecule will rise up to a height of

(A) $8.2\ km$

(B) $12.4\ km$

(C) $10.6\ km$

(D) $15.2\ km$

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Q.15

At constant volume, temperature is increased then

(A) collision on walls will be less

(B) number of collisions per unit time will increase

(C) collisions will be in straight lines

(D) collisions will not change

Q.16

0.014 kg of nitrogen is enclosed in a vessel at a temperature of 27°C. The amount of heat energy to be supplied to the gas to double the rms speed of its molecules is approximately equal to

(A) 6350 J

(B) 7350 J

(C) 8350 J

(D) 9350 J

Q.17

At what absolute temperature T is the root mean square speed of a hydrogen molecule equal to its escape velocity from the surface of the moon? The radius of moon is R, g is the acceleration due to gravity on moon's surface, m is the mass of a hydrogen molecule and k is the Boltzmann constant

(A) $\frac{mgR}{2k}$

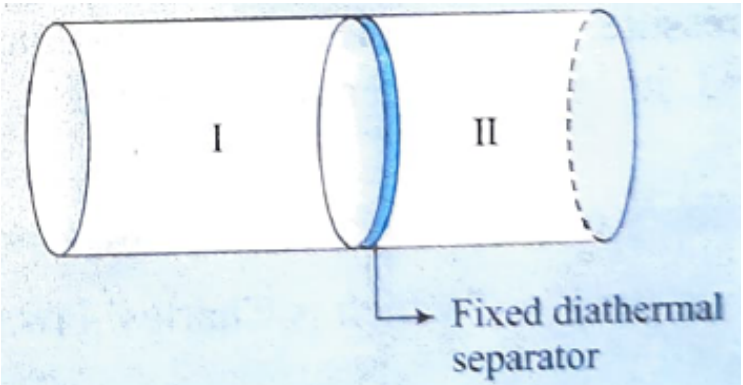
(B) $\frac{2mgR}{k}$

(C) $\frac{3mgR}{2k}$

(D) $\frac{2mgR}{3k}$

Q.18

Figure shows a vessel portioned by a fixed diathermic separator. Different ideal gases are filled in the two parts. The rms speed of the molecules in the left part equal the mean speed of the molecules in the right part. Calculate the ratio of the mass of a molecule in the left part and the mass of a molecule in the right part.



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(A) $\frac{3\pi}{8}$

(B) $\frac{33\pi}{7}$

(C) $\frac{4\pi}{9}$

(D) $\frac{3\pi}{5}$

Q.19

The molecules of a given mass of a gas have root mean square speeds of $100\ \mathrm{ms}^{-1}$ at $27^{\circ}\mathrm{C}$ and 1.00 atmospheric pressure. What will be the root mean square speeds of the molecules of the gas at $127^{\circ}\mathrm{C}$ and 2.0 atmospheric pressure?

(A) $\frac{300}{\sqrt{3}}\ \mathrm{m/s}$

(B) $\frac{200}{\sqrt{3}}\ \mathrm{m/s}$

(C) $\frac{600}{\sqrt{3}}\ \mathrm{m/s}$

(D) $\frac{100}{\sqrt{3}}\ \mathrm{m/s}$

Q.20

At T temperature (in ˚C), the r.m.s velocity of oxygen gas would be half of its value at S.T.P. Given that Pressure remains constant. Then $\frac{\mathrm{T}}{10}$ is

(A) -20.57

(B) -20.75

(C) -20.47

(D) -20.74

Answers & Solutions

Q.1 Answer:

$2 \times 10^{9}\ \mathrm{K}$

Solution:

$\frac{3}{2}\mathrm{k_B T} = \mathrm{K_{av}}$

where $\mathrm{K_{av}}$ is the average kinetic energy of the proton. $\therefore \quad \mathrm{T} = \frac{2\ \mathrm{K_{av}}}{3\mathrm{k_B}}$

$\mathrm{T} = \frac{2 \times 4.14 \times 10^{-14}\ \mathrm{J}}{3 \times 1.38 \times 10^{-23} \mathrm{JK}^{-1}} = 2 \times 10^{9}\ \mathrm{K}$

Q.2 Answer:

965

Solution:

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Root-mean square-velocity is given by

$v_{rms} = \sqrt{\frac{3RT}{M}}$ **i.e.,** $v_{rms} \propto \sqrt{\left(\frac{T}{M}\right)}$

$\therefore \frac{(v_{rms})O_2}{(v_{rms})H_2} = \sqrt{\left[\frac{T_{O_2}}{T_{H_2}} \times \frac{M_{H_2}}{M_{O_2}}\right]}$

$\therefore (v_{rms})\,O_2 = (v_{rms})\,H_2 \times \frac{1}{2} = \frac{1930}{2}$

$= 965\ m/s$

Q.3 Answer:

1

Solution:

For 1 molecule of a gas, $v_{rms} = \sqrt{\frac{3KT}{m}}$

where m **is the mass of one molecule**

For N **molecule of a gas,** $v_1 = \sqrt{\frac{3KT \times N}{m}}$

$\therefore \frac{v_1}{v_2} = 1$

Q.4 Answer:

1.414

Solution:

(1.414) From graph, $T^2\,V =$ **const....(i)**

As we know that $TV^{\gamma-1} =$ **const**

$\Rightarrow VT^{\frac{1}{\gamma-1}} =$ **const....(ii)**

On comparing (1) **and** (2)**, we get** $\Rightarrow \gamma = 3/2$

Also $v_{rms} = \sqrt{\frac{3P}{\rho}}$ **and** $v_{sound} = \sqrt{\frac{P\gamma}{\rho}}$

$\Rightarrow \frac{v_{rms}}{v_{sound}} = \sqrt{\frac{3}{\gamma}} = \sqrt{2} = 1.414$

Q.5 Answer:

$484\ ms^{-1}$

Solution:

Velocity of sound in a gas is $v\sqrt{\frac{\gamma P}{\rho}}$;

rms speed $C = \sqrt{\frac{3P}{\rho}}$

Dividing $\frac{C}{v} = \sqrt{\frac{3}{\gamma}} \Rightarrow C = v\sqrt{\frac{3}{\gamma}}$

$= 332 \times \sqrt{\frac{3}{1.41}} = 484\ m/s$

Q.6 Answer:

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$\sqrt{54/4}$

Solution:

$$v_{rms} = \left[\frac{(2)^2+(3)^2+(4)^2+(5)^2}{4}\right]^{1/2}$$
$$= \sqrt{\left[\frac{54}{4}\right]}$$

Q.7 Answer:

$190.2\ K$

Solution:

If T be the required temperature, then, $v_{rms} - v_p = 200\ m/s$

$\sqrt{\frac{3RT}{M_{He}}} - \sqrt{\frac{2RT}{M_{He}}} = 200\ m/s$

or $\sqrt{T}[\sqrt{3} - \sqrt{2}] = \sqrt{\frac{M_{He}}{R}}(200\ m/s)$

Substituting $M_{He} = 4 \times 10^{-3}\ kg, R = 8.314\ J/$

$mol - K,$

$T = 190.2\ K$

Q.8 Answer:

$2\sqrt{\frac{2\,N+1}{6(\,N+1)}}$

Solution:

$$V_{rms} = \sqrt{\frac{1^2 + 2^2 + \ldots\ldots + N^2}{N}}$$
$$= \sqrt{\frac{N(N+1)(2\,N+1)}{6\,N}}$$
$$V_{mean} = \frac{1 + 2 + \ldots N}{N} = \frac{N(N+1)}{2\,N}$$

Q.9 Answer:

the oxygen is at the higher temperature

Solution:

$v_{rms} = \sqrt{\frac{3RT}{M}}$

Same V_{rms}, implies $\frac{T}{M}$ same. Hence, $T \propto M$ M for oxygen is more than that of hydrogen, therefore, oxygen is at higher temperature than hydrogen.

Q.10 Answer:

H_2

Solution:

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$$\because \quad \mathrm{C}=\sqrt{\frac{3 \mathrm{RT}}{\mathrm{M}}}$$

$$(1930)^2=\frac{3 \times 8.314 \times 300}{\mathrm{M}}$$

$$\mathrm{M}=\frac{3 \times 8.314 \times 300}{1930 \times 1930} \approx 2 \times 10^{-3} \mathrm{~kg}$$

The gas is H_2.

Q.11 Answer:

get doubled

Solution:

$c_{\mathrm{rms}} \propto \sqrt{T}$

As temperature increases from $300 \mathrm{~K}$ to 1200 K that is four times, so, c_{rms} will be doubled.

Q.12 Answer:

$\mathrm{T}_1<\mathrm{T}_2$

Solution:

Higher is the temperature greater is the most probable velocity.

Q.13 Answer:

2v

Solution:

$\frac{(V_{rms})_1}{(V_{rms})_2}=\sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{V}{(V_{rms})_2}=\sqrt{\frac{120}{480}}$

$\Rightarrow \frac{V}{(V_{rms})_2}=\frac{1}{2} \Rightarrow (V_{\mathrm{rms}})_2=2V$

Q.14 Answer:

$12.4 \mathrm{~km}$

Solution:

$v_{\mathrm{rms}}=\sqrt{\frac{3RT}{M}}=\sqrt{\frac{3 \times 8.31 \times 273}{28 \times 10^{-3}}}=493 \mathrm{~m/s}$

Thus $\frac{1}{2} m v_{rms}^2=\frac{mgh}{1+\frac{h}{R}}$. After substituting the values and simplifying, we get $h=12.4 \mathrm{~km}$

Q.15 Answer:

number of collisions per unit time will increase

Solution:

As the temperature increases, the average velocity increases. So, the collisions are faster.

Q.16 Answer:

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9350 J

Solution:

The root mean square speed is related to absolute temperature T as

$c_{\text{rms}} = \sqrt{\frac{3kT}{m}}$

For a given gas, m is fixed. Therefore, $c_{\text{rms}} \propto \sqrt{T}$. Hence in order to double the root mean square speed, the absolute temperature must be increased to four times the initial value. Initial temperature $T_1 = 273 + 27 = 300$ K. Therefore, final temperature $T_2 = 4T_1 = 1200$ K

Since the volume of the vessel is fixed, $\Delta V = 0$. Hence the heat energy supplied to the gas does no work on the gas; it only increases the internal energy of the molecules.

The increase in internal energy is

$\Delta U = nC_v\Delta T$

Since nitrogen is diatomic $C_v = \frac{5R}{2}$. The number of moles of nitrogen in the vessel is

$n = \frac{\text{mass in kg}}{\text{molecular mass}} = \frac{0.014\times10^3}{28} = 0.5$

$\therefore \quad \Delta U = 0.5 \times \frac{5R}{2} \times (1200 - 300)$

$\therefore \quad = \frac{0.5\times5\times8.314\times900}{2} = 9353 \text{ J}.$

So the correct choice is (d)

Q.17 Answer:

$\frac{2mgR}{3k}$

Solution:

The root mean square speed is given by

$v_{\text{rms}} = \frac{\sqrt{3kT}}{m}$

The escape velocity is given by $v_e = \sqrt{2gR}$

For $v_{\text{rms}} = v_e$, we require

$\sqrt{\frac{3kT}{m}} = \sqrt{2gR}$ **or** $T = \frac{2mgR}{3k}$**, which is choice (d).**

Q.18 Answer:

$\frac{3\pi}{8}$

Solution:

The separator is diathermal which means that the temperature of gas on both parts of vessel should be equal. Let this common temperature be T.

Hence, rms speed of a molecule of the gas in left part,

$v_{\text{rms}} = \sqrt{\frac{3kT}{m_1}}$ **...(i)**

Here, m_1 is the mass of a molecule in left part.

Similarly, the mean speed of a molecule of the gas in right part,

$v_{\text{mean}} = \sqrt{\frac{8kT}{\pi m_2}}$ **...(ii)**

Here, m_2 is the mass of a molecule in right part.

As per question, $r_{\text{rms}} = v_{\text{mean}}$

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$\Rightarrow \sqrt{\frac{3kT}{m_1}} = \sqrt{\frac{8kT}{\pi m_2}}$.

Hence, $\frac{m_1}{m_2} = \frac{3\pi}{8}$

Q.19 Answer:

$\frac{200}{\sqrt{3}}\ \mathrm{m/s}$

Solution:

We know that for a given mass of a gas $v_{\mathrm{rms}} = \sqrt{\frac{3RT}{M}}$

where R **is gas constant,** T **is temperature in Kelvin and** M **is molar mass of the gas.**

Clearly for a given gas, $v_{\mathrm{rms}} \propto \sqrt{T}$**, as** R **and** M **are constants.**

Hence, $\frac{(v_{\mathrm{rms}})_1}{(v_{\mathrm{rms}})_2} = \sqrt{\frac{T_1}{T_2}}$

Given, $(v_{\mathrm{rms}})_1 = 100\ \mathrm{m/s}$

$T_1 = 27^{\circ}\mathrm{C} = 27 + 273 = 300\ \mathrm{K}$

$T_2 = 127^{\circ}\mathrm{C} = 127 + 273 = 400\ \mathrm{K}$

Thus, from Eq. (i), $\frac{100}{(v_{\mathrm{rms}})_2} = \sqrt{\frac{300}{400}} = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad (v_{\mathrm{rms}})_2 = \frac{2\times 100}{\sqrt{3}} = \frac{200}{\sqrt{3}}\ \mathrm{m/s}$

Q.20 Answer:

-20.47

Solution:

$C_{r\cdot m.s} \propto \sqrt{T}$

$\therefore \quad \frac{c_{r.m.s1}}{c_{r.m.s_2}} = \sqrt{\frac{T_1}{T_2}}$

$\therefore \frac{c_1}{\frac{c_1}{2}} = \sqrt{\frac{273}{T_2}}$ ... $T_1 = 273\mathrm{K}$ at S.T.P.

$\therefore \quad T_2 = \frac{273}{4}$

$T_2 = 68.25\mathrm{K} = (68.25 - 273)^{\circ}\mathrm{C}$

$\therefore \quad T_2 = -204.75^{\circ}\mathrm{C}$

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